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Optimization II

Introduction to Machine Learning, Lecture 4

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Recap: extension of toolkit

Training Data → ML method: efficiently find $\hat{f} \in F$ → Model $\hat{f}$

ML Scientists design



Function class F	Training loss	Optimization
• linear functions	• squared loss	• closed-form
• nonlinear functions		• gradient methods

Recap: Nonlinear regression

- x: vector of attributes (directly collected from object), $\phi(x)$: vector of features

  where each $\phi_j(x)$ is a separate function of whole vector x,

  but $\phi(x) = \left(\phi_1(x), \dots, \phi_p(x)\right) \in \mathbb{R}^p$ is a vector (of transformed attributes)

- Fixed feature nonlinear function classes : $F_\phi = \left\{f\colon f(x) = \sum_{j=1}^{p} w_j\, \phi_j(x) \text{ for } w \in \mathbb{R}^p\right\}$

  here each $f(x) = \langle w, \phi(x)\rangle$ which is a linear function in $\phi(x)$ but not in x!

- We can use machinery from linear regression to solve regression in nonlinear function spaces by just replacing the data matrix X by the feature matrix Φ

Recap: Steepest descent & gradient descent

- For a general loss function $L(w)$, negative gradient points in steepest descent direction
  - because gradient is "orthogonal to" contour line
  - using linear approximation argument

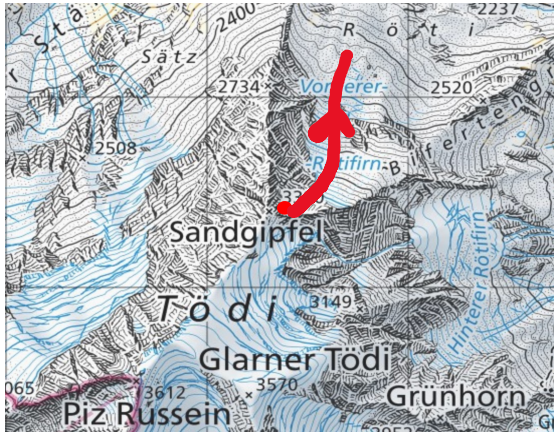
Gradient descent algorithm to minimize $L(w)$

Start at initial w^0

At each step t, $w^{t+1} = w^t - \eta \nabla_w L(w^t), t \leftarrow t+1$ (for simplicity, fixed for all t)

If e.g. $|L(w^t) - L(w^{t-1})| > \epsilon$, continue

Else stop and output $\hat{w} = w^t$



Clarification: Gradient descent convergence to stationary point



For a general function $L(w)$ that has a global minimizer w_{min} .

What does w^t converge to if we let $t \to \infty$ and choose small enough stepsize η? It depends!

Reasoning is delicate:

- Note: Here we don't assume a stopping criterion at a threshold

  (with which you would output an ϵ-stationary point)

- The point is: The sequence of iterates might not converge (mathematically)

  even though gradient norms convergence to zero

- For a sufficient condition for GD to converge to a local minimum, see e.g. Theorem 5.18

  nontrivial to characterize necessary & sufficient assumptions for convergence to stationary points.

BONUS

Your question: Comparing different stopping criteria

- In the slides we randomly picked as concrete metric to measure "not too much change" in last iteration:

$$|L(w^t) - L(w^{t-1})| \leq \epsilon. \quad (1)$$

  Other candidates were $\left||\nabla L(w^{t-1})|\right| \leq \epsilon$ (2) or equivalently $\left||\mathrm{w}^t - w^{t-1}|\right| \leq \epsilon$

- For (2) by definition, you directly see that you may end up at (close to) stationary points
- (1) implies (2) for locally nicely behaved functions (e.g. locally smooth functions (Lipschitz gradient))
- Some advantages of choosing (1) (to be understood later)
  - More robust if the gradients that you use are noisy/stochastic (like later this lecture)
  - For neural networks: Same function (and loss) can be represented by different parameter scaling that change gradients arbitrarily
- In reality, for optimization in ML, people neither use the training loss nor gradients (next week)

Plan for today

- Gradient-based methods
  - convergence of gradient descent for linear regression
  - convergence speed and variants of GD
  - stochastic gradient method
- Convexity and optimality of stationary points

Convergence to optimal w_{min} for linear regression

(same minimizer as $\frac{1}{n}\left|\left|y - Xw\right|\right|^2$ but simpler notation)

- To minimize rescaled $L(w) = \frac{1}{2}\left|\left|y - Xw\right|\right|^2$, use gradient descent update: $w^{t+1} = w^t - \eta \nabla L(w^t)$
- Using chain rule (or check manually), gradient at any w reads $\nabla L(w) = -X^\top(y - Xw)$
- We now show that indeed $w^t \to w_{min}$ and $||w^t - w_{min}||_2 \leq \rho^t ||w^0 - w_{min}||_2$
- And we obtain $w^{t+1} = w^t - \eta \nabla L(w^t) = w^t + \eta\, X^\top(y - Xw^t)$
- Furthermore, remember from Lecture 2 that the global minimum w_{min} satisfies $X^\top X w_{min} = X^\top y$

$$w^{t+1} - w_{min} = w^t + \eta\, X^\top(y - Xw^t) - w_{min}$$

$X^\top X w_{min} = X^\top y$ →

$$= w^t - w_{min} + \eta(X^\top X w_{min} - X^\top X w^t)$$

$$= (I - \eta X^\top X)(w^t - w_{min})$$

…derivation skipped in class

Convergence to optimal w_{min} for linear regression

(same minimizer as $\frac{1}{n}\left||y - Xw|\right|^2$ but simpler notation)

- To minimize rescaled $L(w) = \frac{1}{2}\left||y - Xw|\right|^2$, use gradient descent update: $w^{t+1} = w^t - \eta \nabla L(w^t)$
- Using chain rule (or check manually), gradient at any w reads $\nabla L(w) = -X^\top(y - Xw)$
- Further using $X^\top X w_{min} = X^\top y$, you can check that $w^{t+1} - w_{min} = (I - \eta X^\top X)(w^t - w_{min})$ (see previous slide)

$$\left||w^{t+1} - w_{min}|\right|_2 = \left||(I - \eta X^\top X)(w^t - w_{min})|\right|_2 \leq \underbrace{\left||I - \eta X^\top X|\right|_{op}}_{\rho} \left||w^t - w_{min}|\right|_2$$

$$\leq \rho \cdot \rho \left||w^{t-1} - w_{min}|\right|_2 \leq \rho^{t+1}\left||w^0 - w_{min}|\right|_2$$

Recall definition of operator norm for any matrix A: $\left||A|\right|_{op} = \sup_z \frac{\left||Az|\right|_2}{\left||z|\right|_2}$

Convergence to optimal w_{min} for linear regression

ρ^t initial distance (independent of t)

$$||w^t - w_{min}||_2 \leq \left[\left||I - \eta X^\top X|\right|_{op}\right]^t ||w^0 - w_{min}||_2$$

- If $\rho < 1$, then we have contraction $||w^t - w_{min}||_2 < ||w^0 - w_{min}||_2$ leading to convergence

  and we say the distance converges linearly / exponentially to 0 as $t \to \infty$

- When is $\rho < 1$? We can choose η accordingly:

  In particular, we can derive $\rho = \max\{|1 - \eta\lambda_{min}(X^\top X)|, |1 - \eta\lambda_{max}(X^\top X)|\}$*

  → If $\eta \leq 2/\lambda_{max}(X^\top X)$ and $X^\top X$ full-rank, then $\rho < 1$ (see details in Theorem 5.7., lecture notes)

*uses that for any real symmetric matrix A, the operator norm is $\left||A|\right|_{op} = \max\{|\lambda_{max}(A)|, |\lambda_{min}(A)|\}$

Convergence to optimal w_{min} for linear regression

ρ^t initial distance (independent of t)

$$||w^t - w_{min}||_2 \leq \left[\left||I - \eta X^\top X|\right|_{op}\right]^t ||w^0 - w_{min}||_2$$

- The optimal (constant) learning rate is $\eta_{opt} = \frac{2}{\lambda_{max}(X^\top X)+\lambda_{min}(X^\top X)}$ leading to $\rho_{min} = \frac{\lambda_{max}-\lambda_{min}}{\lambda_{max}+\lambda_{min}} < 1$

  (see proof in Section 5.3.1. in lecture notes)

- Obviously one cannot quickly compute the max and min eigenvalue

  → often η is a hyperparameter to choose (see model selection lectures next week!)

we now use short notation $\lambda_{max} = \lambda_{max}(X^\top X)$ and $\lambda_{min} = \lambda_{min}(X^\top X)$

Plan for today

- Gradient-based methods
  - convergence of gradient descent for linear regression
  - convergence speed and variants of GD
  - stochastic gradient method
- Convexity and optimality of stationary points

GD with constant stepsize η:
$w^{t+1} = w^t - \eta \nabla L(w^t)$

Conditioning and convergence speed

We are minimizing $L(w) = \frac{1}{2}\left||y - Xw|\right|^2$ and assume $X^\top X = \begin{bmatrix} \lambda_{min} & 0 \\ 0 & \lambda_{max} \end{bmatrix}$



Gradient descent needs fewer iterates to converge, using optimal step size η_{opt}, if

(A) $\lambda_{max} \gg \lambda_{min}$ (B) $\lambda_{max} \approx \lambda_{min}$

Solution: (B). Two ways to see:

Using calculation: optimal step size → $\rho_{min} = \frac{\lambda_{max} - \lambda_{min}}{\lambda_{max} + \lambda_{min}} = \frac{\frac{\lambda_{max}}{\lambda_{min}} - 1}{\frac{\lambda_{max}}{\lambda_{min}} + 1}$ (minimal when $\frac{\lambda_{max}}{\lambda_{min}}$=1))

Or drawing visually as in the next slides

Visualization: Contour lines for linear regression in 2d

- Recall the loss $L(w) = \frac{1}{2}\left||y - Xw|\right|^2$ and closed-form optimum w_{min} satisfying $X^\top y = X^\top X w_{min}$
- Assume for simplicity $X^\top X = \begin{bmatrix} \lambda_{min} & 0 \\ 0 & \lambda_{max} \end{bmatrix}$, can then by completing the square that

  $L(w) = \left||X(w - w_{min})|\right|^2 + C$ where C is a term that does not depend on w (see details next slide)

- Hence, contour lines {w: $L(w) = c$} are ellipses around w_{min} w/ width $\propto \frac{1}{\sqrt{\lambda_{min}}}$ and height $\propto \frac{1}{\sqrt{\lambda_{max}}}$

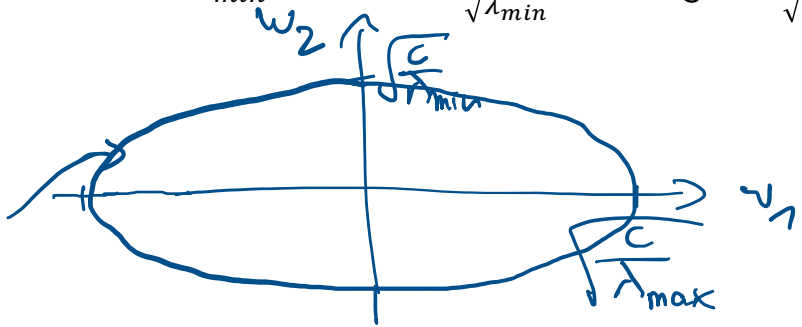
For $w_{min} = 0$

For simplicity

$\{w: w^\top X^\top X w = c\}$

$\begin{bmatrix} \lambda_{min} & 0 \\ 0 & \lambda_{max} \end{bmatrix}$

$w_1 \lambda_{min}^2 + w_2 \lambda_{max}^2 = c$



BONUS

Visualization: Contour lines for linear regression in 2d

- For linear regression we minimize the loss $L(w) = \frac{1}{2} \left|\left|y - Xw\right|\right|^2$
- Assuming for simplicity $X^\top X = \begin{bmatrix} \lambda_{min} & 0 \\ 0 & \lambda_{max} \end{bmatrix}$
- contour lines: w that have the same value, i.e. $\frac{1}{2}[w^\top X^\top X w - 2w^\top X^\top y + y^\top y] = c$

by normal equations and optimality of w_{min}

$\Leftrightarrow \frac{1}{2}[w^\top X^\top X w - 2w^\top X^\top X w_{min} + y^\top y] = \frac{1}{2}\left[||Xw - Xw_{min}||^2 - \left|\left|Xw_{min}\right|\right|^2 + y^\top y\right] = c$

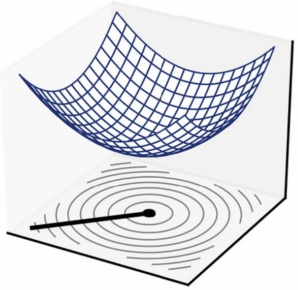
→ by completing the square we get $\lambda_{min}\left(w_{[1]} - w_{min,[1]}\right)^2 + \lambda_{max}\left(w_{[2]} - w_{min,[2]}\right)^2 = \tilde{c}$

depends on X, y, c

→ Ellipses around w_{min} with width $\propto \frac{1}{\sqrt{\lambda_{min}}}$ and height $\propto \frac{1}{\sqrt{\lambda_{max}}}$

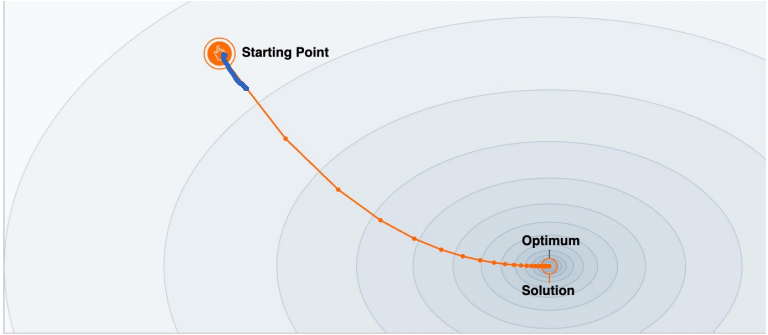
GD with constant stepsize η:
$w^{t+1} = w^t - \eta \nabla L(w^t)$

Well-conditioned landscape…

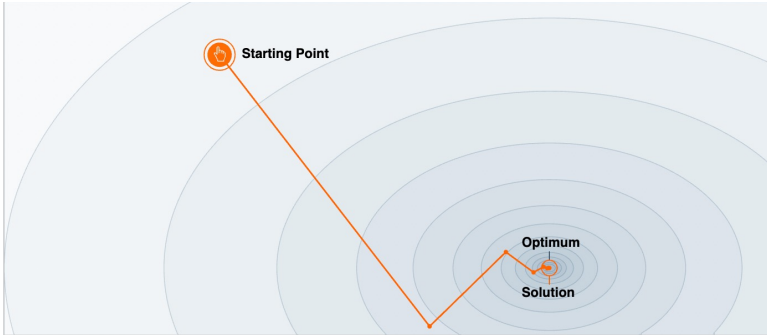


When $\lambda_{max} \approx \lambda_{min}$ (well-conditioned)

- contour lines are ellipses close to spheres
  → similar steepness in all directions
- gradient descent converges fast and
- can take pretty big steps!



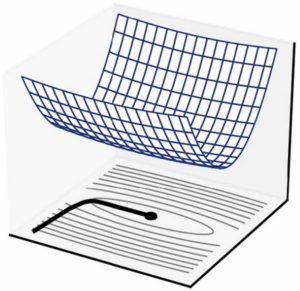
small stepsize



large stepsize

GD with constant stepsize η:
$w^{t+1} = w^t - \eta \nabla L(w^t)$

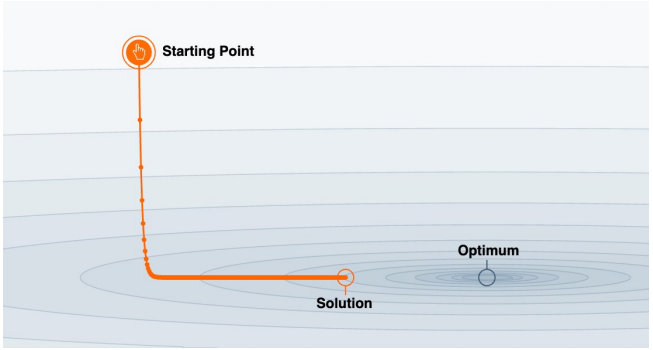
… vs. "ill-conditioned" landscape



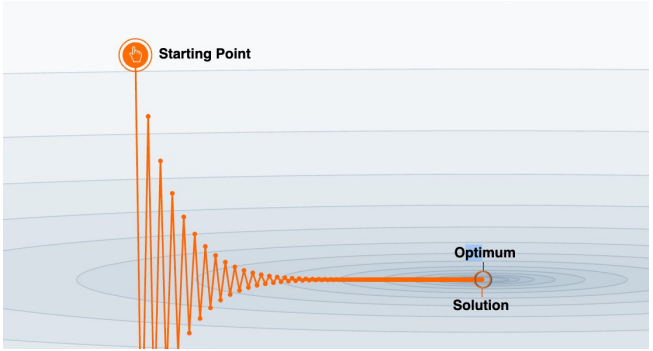
When $\lambda_{\max} \gg \lambda_{min}$ are very different:

- contour lines/ellipses are "stretched"
  → there are very steep and very flat areas
- for small step size: slow in flat area
- for larger step size: large oscillations

→ even with the best stepsize it takes forever to converge!



small stepsize



large stepsize

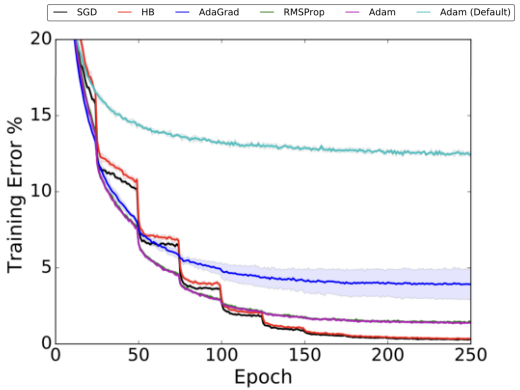
Source: https://fa.bianp.net/teaching/2018/COMP-652/

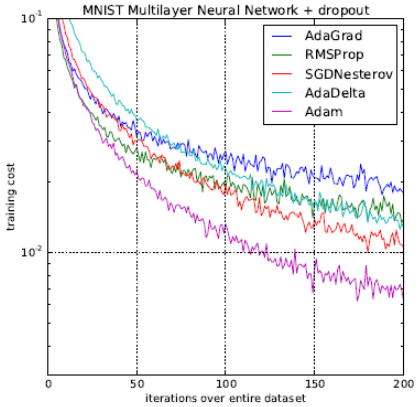
There's a whole zoo of optimization methods…

For deep learning, there are many gradient-based methods (different lines) people usually try out (don't parse the plots in detail)

→ now want to have a brief overview how they differ from gradient descent







(Here they plot classification training loss/error on CIFAR10, MNIST image datasets)

Goal I: Speeding up gradient descent

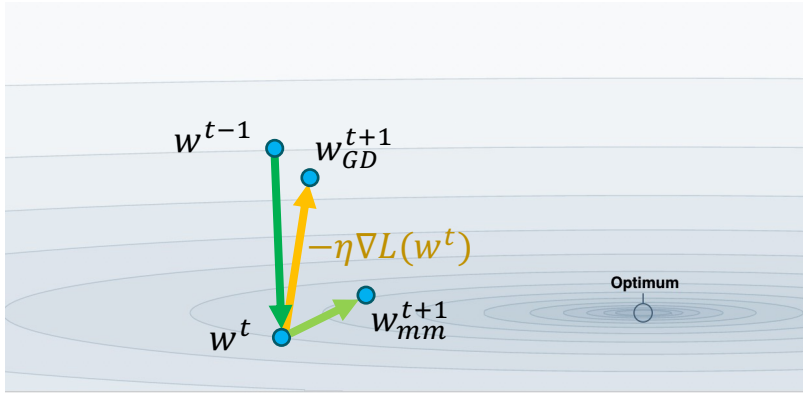
Goal: Large steps in flat areas, small steps in high curvature ones, dampened oscillations

- momentum/accelerated methods: add weighted previous step to neg. gradient

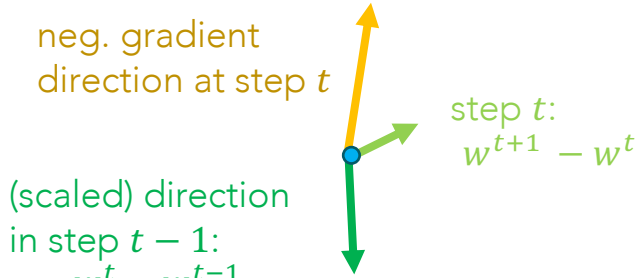
$$w^{t+1} - w^t = \beta\,(w^t - w^{t-1}) - \eta \nabla_w L(w^t)$$ for some $\beta \in \mathbb{R}$

(bonus: discrete approximation of dampened harmonic oscillator)

how oscillations are dampened for ill-conditioned linear regression



At time step t:



Nice overview blog posts: https://distill.pub/2017/momentum/; https://www.ruder.io/optimizing-gradient-descent/index.html

Goal I: Speeding up gradient descent

Goal: Large steps in flat areas, small steps in high curvature ones, dampened oscillations

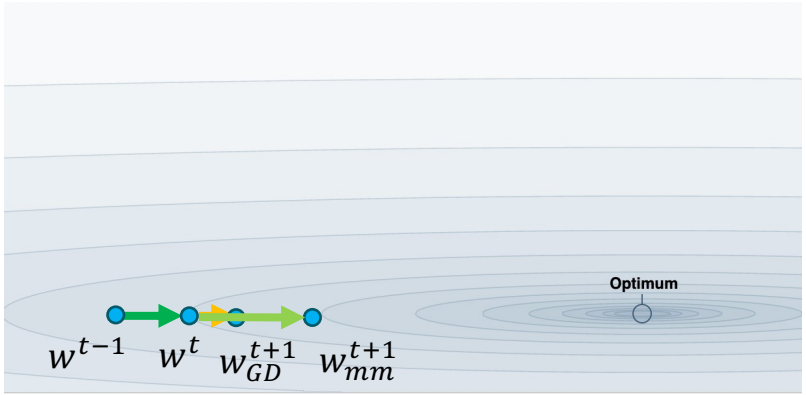
- momentum/accelerated methods: add weighted previous step to neg. gradient

$$w^{t+1} - w^t = \beta\,(w^t - w^{t-1}) - \eta \nabla_w L(w^t)$$

for some $\beta \in \mathbb{R}$

(bonus: discrete approximation of dampened harmonic oscillator)

speed up in flat areas for ill-conditioned linear regression



At time step t:

neg. gradient direction at step t

(scaled) direction in step $t-1$: $(w^t - w^{t-1})$

step t: $w^{t+1} - w^t$

Nice overview blog posts: https://distill.pub/2017/momentum/; https://ruder.io/optimizing-gradient-descent/index.html

BONUS

Goal I: Speeding up gradient descent

Goal: Large steps in flat areas, small steps in high curvature ones, dampened oscillations

- momentum/accelerated methods: add weighted previous step to neg. gradient

$$w^{t+1} - w^t = \beta\,(w^t - w^{t-1}) - \eta \nabla_w L(w^t) \qquad \text{for some } \beta \in \mathbb{R}$$

(bonus: discrete approximation of dampened harmonic oscillator)

- adaptive methods (AdaGrad, RMSProp, Adam etc.): Different stepsize for different entries w_i

intuitively: the elements i which already changed a lot, have smaller stepsize

$$w_i^{t+1} = w_i^t - \frac{\eta}{\sqrt{\text{previouschange}_i + \gamma}} \frac{\partial L}{\partial w_i}(w^t)$$

- 2nd order methods (Quasi-Newton, L-BFGS): https://fa.bianp.net/teaching/2018/eecs227at/newton.html

Nice overview blog posts: https://distill.pub/2017/momentum/; https://www.ruder.io/optimizing-gradient-descent/index.html

Plan for today

- Gradient-based methods
  - convergence of gradient descent for linear regression
  - convergence speed and variants of GD
  - stochastic gradient method
- Convexity and optimality of stationary points

Goal II: Saving memory → stochastic gradient method

Recall: For ML, the losses we try to minimize are averages over pointwise losses over all training data

Gradient descent algorithm

Start at initial w^0

Until $w^{t+1} - w^t$ small, repeat:

Update $w^{t+1} \leftarrow w^t - \eta \ \nabla_w L(w^t)$

Output $\hat{w} = w^T$

average over gradients at all points

$\nabla_w L(w) = \frac{1}{n}\sum_{i=1}^{n} \nabla_w \ell(y_i, f_w(x_i))$

S(tochastic) G(radient) D(escent) algorithm

Start at initial w^0

Until $w^{t+1} - w^t$ small, repeat:

Update $w^{t+1} \leftarrow w^t - \eta \ \nabla_w L_S(w^t)$

Output $\hat{w} = w^T$

average over gradients on random subset of points $S \subset [1, \dots, n]$ (minibatch SGD* with batchsize $|S|$)

$\nabla_w L_S(w) = \frac{1}{|S|}\sum_{i \in S} \nabla_w \ell(y_i, f_w(x_i))$

Costly! memory: require $O(nd)$ values to compute
computation: $O(n \times$ cost to compute $\nabla l)$ per update.

*originally "SGD" referred to when $|S| = 1$, nowadays it also refers to minibatch SGD with batch size >1

Goal II: Saving memory → stochastic gradient method

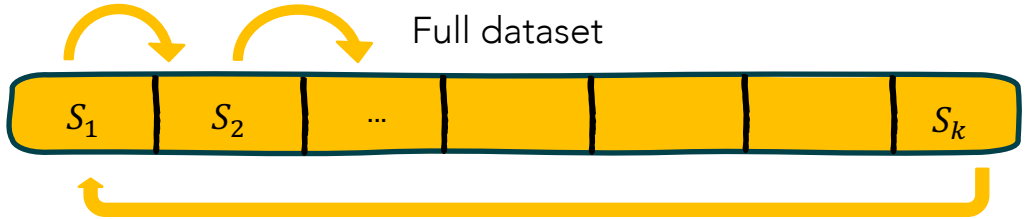
- For any w, if we sample S independently and uniformly with replacement, the expectation

  $\mathbb{E}[\nabla_w L_S(w)] = \mathbb{E}\frac{1}{|S|}\sum_{i\in S}\nabla_w \ell(y_i, f_w(x_i)) = \nabla_w L(w)$ over the draw of S is equal to the full gradient!

- In practice, S_t for every iteration is pre-set and we cycle through the minibatches in the data

Iteration/update:
$w^1 = w^0 - \eta\nabla_w L_{S_1}(w^0)$

Iteration/update:
$w^2 = w^1 - \eta\nabla_w L_{S_2}(w^0)$

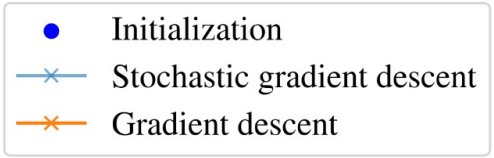


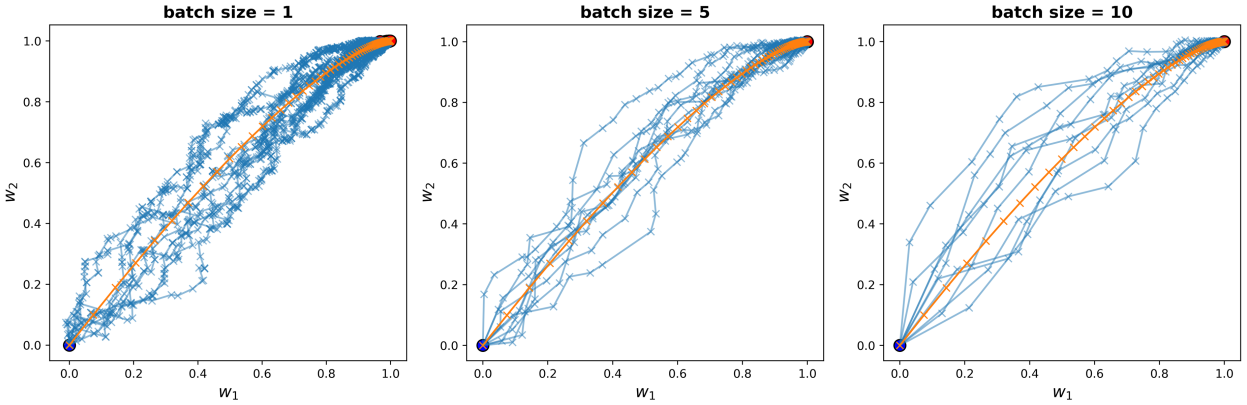
SGD - The effect of batch size

GD/SGD run on the 2-d linear regression objective with $n = 50$ training samples,

with different batch size $|S|$ but same learning rate

- Blue curves here correspond to SGD paths when cycling through a pre-defined partition (into sets of the corresponding batch size) multiple times as in the previous slide
- Each curve corresponds to a different partition of the data (or equivalently, permutations of the data)



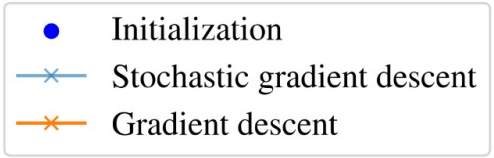


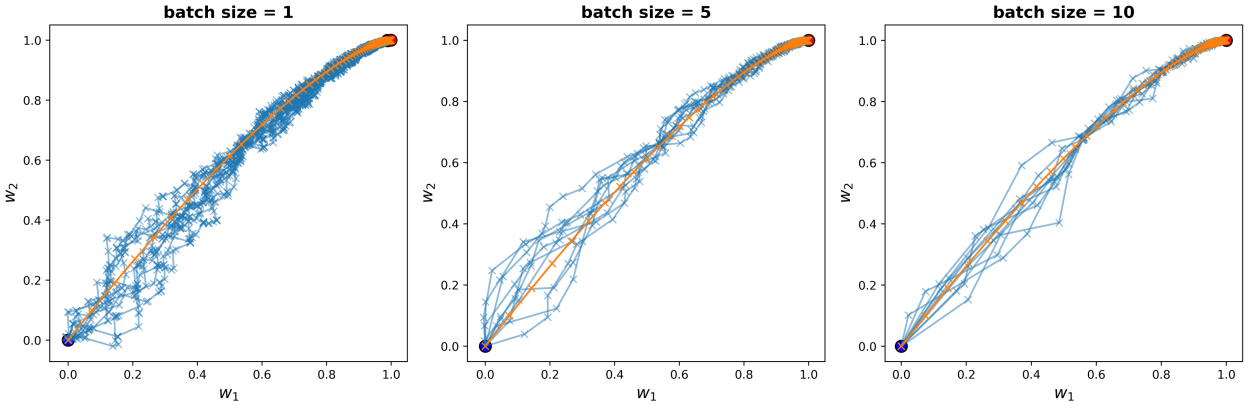
SGD - The effect of batch size - similar for resampling

GD/SGD run on the 2-d linear regression objective with $n = 50$ training samples,

with different batch size $|S|$ but same learning rate

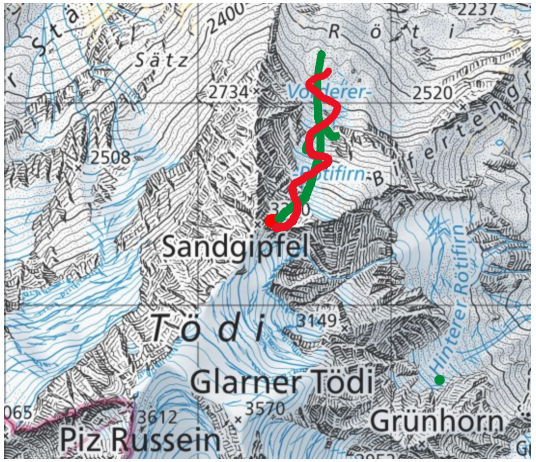
- Blue curves here corresponds to SGD paths when resampling a batch of size $|S|$ in each iteration
- Each curve corresponds to running SGD with different random seeds



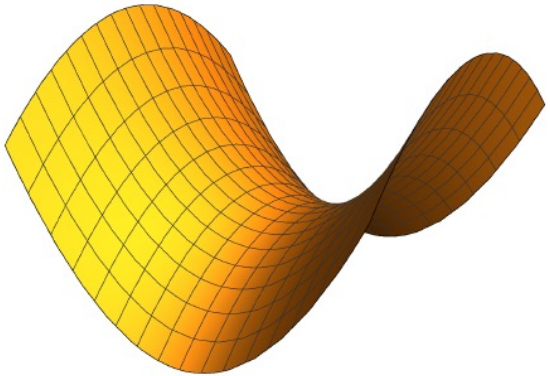


Summary: SGD vs. GD - The effect of batch size

Effects of reducing gradient of the full batch of data to a subset S, beyond computational advantage:



batchsize ↓

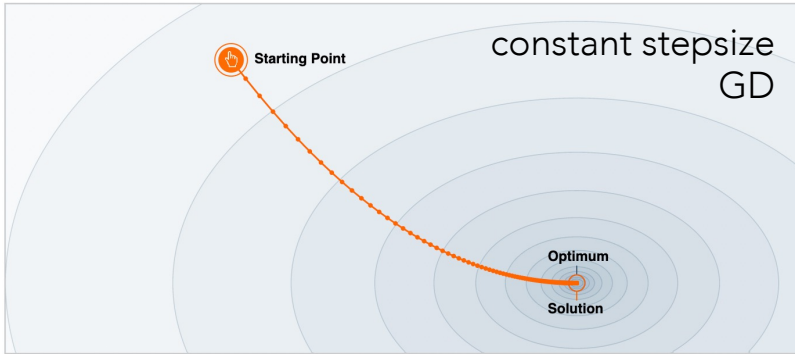


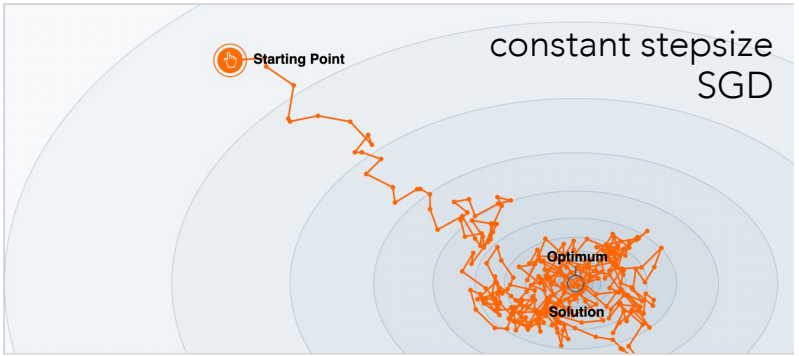
GD can get stuck
in stationary points

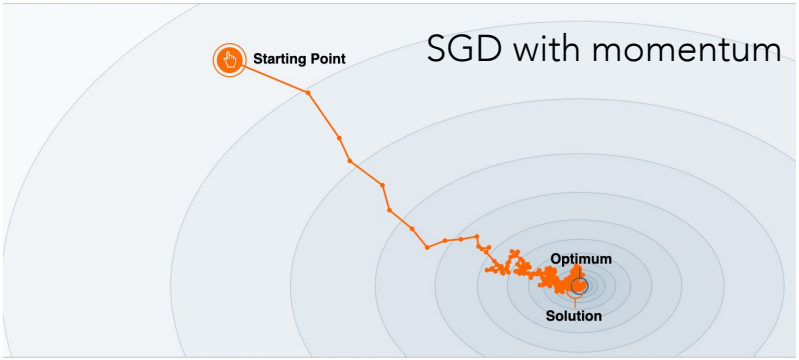
variance ↑ 😢

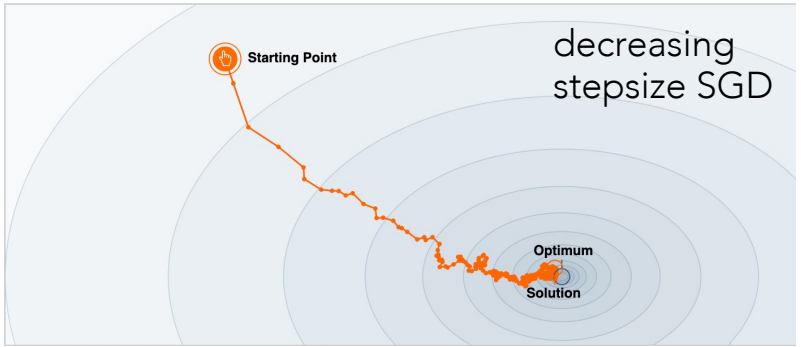
chance to escape saddle points ↑ 😀

GD vs. variants of SGD









Source: https://fa.bianp.net/teaching/2018/eecs227at/stochastic_gradient.html

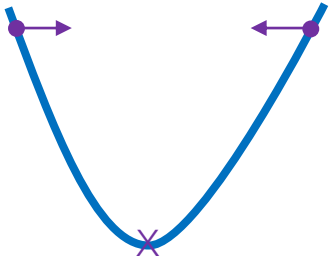
Plan for today

- Gradient-based methods
- Convexity and optimality of stationary points
  - Convexity
  - Strong convexity
  - Optimality of stationary points

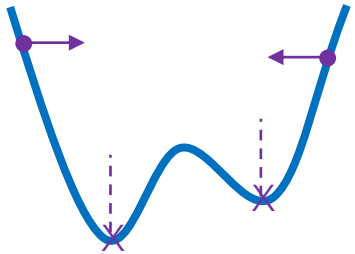
When does GD converge to local/global minima?

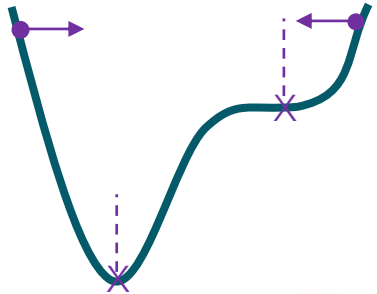
Consider again $d = 1$ and three different functions $L(w)$ that have stationary points. Reminder:

- Local minimum w_{loc}: $L(w_{loc}) \leq L(w)$ for all w in a neighborhood of w_{loc}
- Global minimum w_{glob}: $L(w_{glob}) \leq L(w)$ for all w in the domain of L



Local = global minimum 😀





Might find stationary point with $\nabla_w L(w) = 0$ but possibly not global minimum 😢

Recap: Stationary points and minima

Gradient descent may converge to a stationary point, but we want to get to minimum!

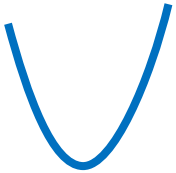
Remember: For differentiable losses,

w is local minimum	always →	w is stationary point
w is global minimum	always →	w is local minimum
$L(w) < L(v)$ for all $v \neq w \in \mathbb{R}^d$	always →	w is global minimum

When can we say something about the reverse? → General notions of convexity

Short primer on convexity

A function L is convex if and only if for all w, v in its domain it holds that:



- 0-th order condition: for all $\lambda \in [0,1]$ $L(\lambda w + (1-\lambda)v) \leq \lambda\, L(w) + (1-\lambda)L(v)$

  true function connecting any w, v $\leq$ linear function connecting two points $L(w), L(v)$



- 1st order condition: $L(v) \geq L(w) + \nabla L(w)^{\top}(v-w)$

  value at any v of linear approximation at any w $\leq$ true value at v

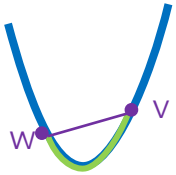
- 2nd order condition: Hessian $\nabla^2 L(w) \succcurlyeq 0$

  non-negative curvature at every w

Note: If these conditions only hold for all w, v in a neighborhood, we say L is locally convex

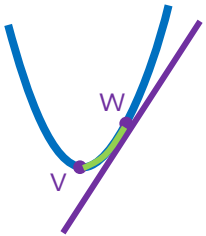
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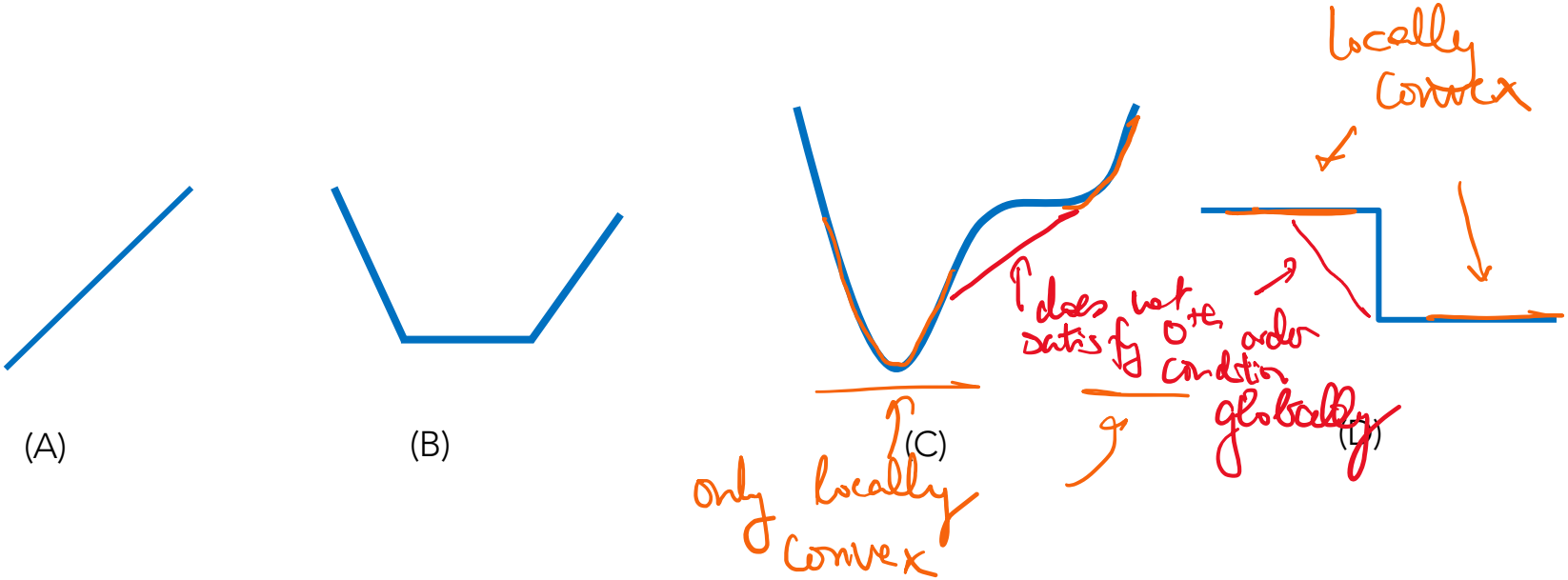
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- 2nd order condition: Hessian $\nabla^2 L(w) \succcurlyeq 0$

  non-negative curvature at every w

Note: If these conditions only hold for all w, v in a neighborhood, we say L is locally convex

Quiz: Which function is convex?



(A)

(B)

(C)

(D)

Operations that preserve convexity

We care about the training loss $L(w) = \frac{1}{n}\sum_{i=1}^{n} \ell(y_i, f_w(x_i))$ and want to check convexity fast. You can either:

- manually check for the 0th to 2nd order conditions,
- look at a list of standard convex functions and show that it is obtained using a known convex function and some operations that preserve convexity:
  - The sum $\alpha\, f + \beta\, g$ is convex for any $\alpha, \beta \geq 0$ if f, g are convex
  - $f \circ g$ *is* convex if f convex and g affine OR f non-decreasing & convex and g convex
  - $h(x) = \max\{f(x), g(x)\}$ is convex (pointwise maximum) if f, g are convex

notation: $f \circ g(x) = f(g(x))$

Discuss with your neighbor…



1. Let's pick the square loss $\ell\big(y, f_w(x)\big) = \big(y - f_w(x)\big)^2$. Which of the following statements are true?

(a) $L(w)$ is convex (b) $L(w)$ is convex for $f_w(x) = w^\top x$

(c) $L(w)$ is convex for $f_w(x) = \mathrm{relu}(\mathrm{w}^\top x) \coloneqq \max(0, w^\top x)$

Discuss for 8 min and answer both questions on eduApp

2. Noting the convexity conditions $L(v) \geq L(w) + \nabla L(w)^\top (v - w)$

Which of the following statements are true (don't cheat and look at the next slide!):

(A) All convex functions have only one local minimum (if it has one) that is a global minimum

(B) For convex functions, all stationary points are local minima

(C) For convex functions, all stationary points are global minima

Operations that preserve convexity



1. Let's pick the square loss $\ell\big(y, f_w(x)\big) = \big(y - f_w(x)\big)^2$. Which of the following statements are true?

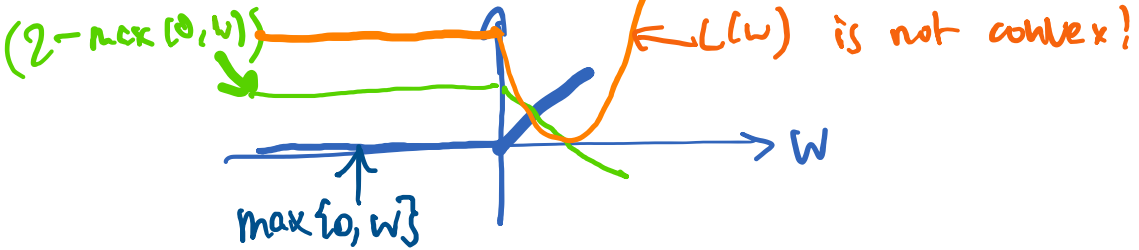
(a) $L(w)$ is convex (b) $L(w)$ is convex for $f_w(x) = w^\top x$

(c) $L(w)$ is convex for $f_w(x) = \mathrm{relu}(\mathrm{w}^\top x) \coloneqq \max(0, w^\top x)$

Solution: (b). In particular, (c) is not, because

even though f_w is convex, it's composed with $\mathrm{f(v)} = (y - v)^2$, which is convex but not a non-decreasing function of v and hence the composition not necessarily convex by our rules! Simple counterexample:

Consider 1-d example where $x = 1, y = 2$. Then we can plot $L(w) = (2 - \max(0, w))^2$



Convexity and stationary points



Noting the convexity conditions $L(v) \geq L(w) + \nabla L(w)^{\mathsf{T}} (v - w)$

Which of the following statements are true (don't cheat and look at the next slide!):

(A) All convex functions have only one local minimum that is a global minimum

(B) For convex functions, all stationary points are local minima

(C) For convex functions, all stationary points are global minima

Solution: (B) and (C) are true immediately if you plug in $\nabla L(w) = 0$ in first order condition

↳ $L(v) \geq L(w)$ $\forall v, w$ $\Leftrightarrow$ w is global minimum

Stationary points and minima revisited

A more complete picture for differentiable functions

w is local minimum

$L(w) \leq L(v)$ for all v in neighborhood of w

always →

← if convex

w is stationary point

$\nabla L(w) = 0$

w is global minimum

$L(w) \leq L(v)$ for all v in the domain of L

always →

← if convex

w is local minimum

w is unique global minimum

$L(w) < L(v)$ for all $v \neq w \in \mathbb{R}^d$

always →

← ?

≙There is a unique global minimum

w is global minimum

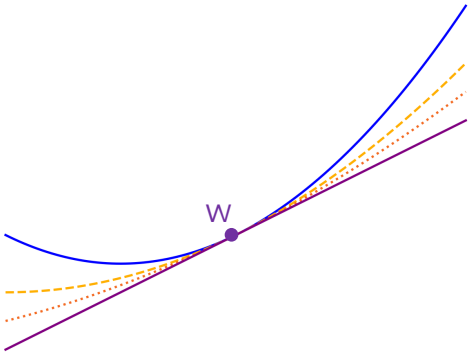
➡ useful concept of strong convexity

Short primer on **strong** convexity

Strongly convex functions are "at least as convex as quadratics". Formally we say:

A function L is strongly convex if and only if for all v, w in its domain it holds that

- 0th order condition: $L(w) - \frac{m}{2}||w||^2$ is convex for some $m > 0$.
- 1st order condition (if and only if for all w exists $m > 0$): $L(v) \geq \overbrace{L(w) + \nabla L(w)^\top (v - w)}^{\text{linear approximation}} + \overbrace{\frac{m}{2}||v - w||^2}^{\text{quadratic lower bound}}$



$\Leftrightarrow$ there exists some quadratic that touches $L(w)$ at w that at any v has value $\leq$ true value at v

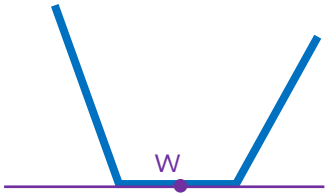
$\Rightarrow$ value at any v of linear approximation $<$ true value at v

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convex but not strongly convex!

$\Leftrightarrow$ there exists some quadratic that touches $L(w)$ at w that at any v has value $\leq$ true value at v

$\Rightarrow$ value at any v of linear approximation $<$ true value at v

Short primer on **strong** convexity

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- 2nd order condition (if and only if for all w some $m > 0$): Hessian $\nabla^2 L(w) \succcurlyeq mI$

  (strictly) positive curvature at every w, i.e. no "flat" areas

Notes:

- if one of those conditions are true for some m, we say L is m −strongly convex
- they don't hold for all w in the domain of L but only i neighborhood, L is locally strongly convex

Stationary points and minima revisited

A more complete picture for differentiable functions

w is local minimum ⟷ w is stationary point (always; if convex)

w is global minimum ⟷ w is local minimum (always; if convex)

$L(w) < L(v)$ for all $v \neq w \in \mathbb{R}^d$ ⟷ w is global minimum (always; if strongly convex*)

The global minimum is unique

In the tutorial, we'll revisit the number of minima from that perspective for linear regression…

*proven by simple contradiction in lecture notes Theorem 5.16.

References

- Lecture notes Chapter 5 (will have minor updates)
- Picture source and animation
  - https://distill.pub/2017/momentum/
  - https://fa.bianp.net/teaching/2018/COMP-652/gradient_descent.html
    https://fa.bianp.net/teaching/2018/eecs227at/stochastic_gradient.html
- More on convexity: Boyd & Vandenberghe “Convex optimization”
- More theoretical foundations for optimization:

  Optimization for data science course at D-INFK